Hazem Masarani

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Graduate researcher focused on LLM inference, distributed systems, and scalable machine learning infrastructure.

Research

PACELab, Stony Brook University 2025 – Present
Graduate Researcher Stony Brook, NY

- Designed and implemented tensor-parallel inference for Mamba large language models using PyTorch Distributed and NCCL
- Extended the vLLM inference engine with custom Mamba execution paths for distributed inference
- Benchmarked communication strategies to evaluate scalability, throughput, and numerical correctness across multiple GPUs
- Conducting research with Prof. Anshul Gandhi on efficient LLM serving and distributed AI systems

Experience

Al-Madinah School Sep 2024 – Aug 2025
Computer Teacher Queens, NY

- Delivered classroom instruction on computer fundamentals, digital literacy, and introductory computer science topics

Etisalat e& Jan 2024 – Sep 2024
Senior Data Automation & Network Data Engineer Egypt (Hybrid)

- Built Python automation systems for large-scale telecom network monitoring and operational analytics
- Designed SQL data pipelines and interactive dashboards used by engineering teams

Etisalat e& Jul 2023 – Jan 2024
Junior Data Automation & Network Data Engineer Egypt (Hybrid)

- Developed Python automation tools for monitoring telecom network performance
- Maintained SQL ETL pipelines supporting enterprise reporting systems

Projects

Tensor Parallelism State Space Models 2026
Research Project

- Implemented tensor parallelism for Mamba language models using PyTorch Distributed and NCCL
- Benchmarked communication strategies and validated numerical correctness across multiple GPUs

vLLM Mamba Extension 2026 – Present
Research Project

- Modified the vLLM inference engine to support custom Mamba execution paths
- Investigated distributed inference architecture and optimized the model execution flow

DC3 Skew Algorithm Implementation 2026
Algorithms and Performance Benchmarking Project

- Implemented the linear-time DC3 (Skew) suffix-array construction algorithm in C++ and Python

Education

Stony Brook University 2025 – Present

M.S. in Data Science | GPA: 3.96 / 4.00 Stony Brook, NY

- Research Advisor: Prof. Anshul Gandhi

Alexandria University 2017 – 2022

B.Sc. in Computer & Communication Engineering | GPA: 3.47 / 4.00 Alexandria, Egypt

- Graduation Project: Online Multi-User IDE with Server-Side Code Execution (A+)

SKILLS

Programming: Python, Java, JavaScript, SQL, C/C++, R

AI / Machine Learning: PyTorch, TensorFlow, Scikit-learn, Mamba, vLLM, Pandas, NumPy

Distributed Systems: PyTorch Distributed, NCCL, Tensor Parallelism, Multi-GPU Inference

Backend & Databases: Django, REST APIs, PostgreSQL, MySQL, MS SQL Server, MongoDB

Developer Tools: Docker, Git, Linux, Plotly, Power BI

Core Skills: Data Structures, Algorithms, Problem Solving, Distributed Computing